

The diagram shows a two-stage DC-DC conversion circuit. The first stage, U2 (LM2575-3.3WU-TR), is a 12V to 5V step-down converter. It takes a 12V input and provides a 5V output. The second stage, U3 (LM2592HVS-5.0), is a 5V to 3.3V step-down converter. It takes the 5V output from U2 and provides a 3.3V output. The circuit includes various passive components like capacitors, inductors, and diodes, and is powered by a 12V source.

The diagram shows the external PCB connections. On the left, a +5V supply and ground are connected to three PWM signals: PWM1, PWM2, and PWM3. These signals are connected to the 1, 2, and 3 pins of three connectors: J4 (SERVO1_CONN), J5 (SERVO2_CONN), and J6 (SERVO3_CONN). On the right, four servos (M2, M3, M4) are connected to the same PWM signals. Each servo is connected to the 1, 2, and 3 pins of a connector: J8 (SERVO1_CONN), J9 (SERVO2_CONN), and J10 (SERVO3_CONN). The servos are labeled M2 SER0039 5V, M3 SER0039 5V, and M4 SER0039 5V.

The diagram shows the following components and connections:

- SW1 SW_Push**: A push button connected to the SW1 pin of the ESP32-S3-WROOM-1 module.
- R6 10k 0.25W**: A pull-down resistor connected between the SW1 pin and ground.
- C11 4.7uF 25V**: A capacitor connected between the +3.3V supply and ground.
- U4 ESP32-S3-WROOM-1**: The main microcontroller module.
- J14 USB_Micro**: A USB Micro module connected to the ESP32-S3-WROOM-1 module.
- J15 FEM_HEADER_01x13**: A 13-pin female header connected to the ESP32-S3-WROOM-1 module.

Pin Connections:

- SW1 SW_Push**: Pin 1 to GND, Pin 2 to SW1 pin of U4.
- R6 10k 0.25W**: One end to SW1 pin of U4, other end to GND.
- C11 4.7uF 25V**: One end to +3.3V supply, other end to GND.
- U4 ESP32-S3-WROOM-1**:
 - EN pin to +3.3V supply.
 - TXD0 pin to TX pin of J14.
 - RXD0 pin to RX pin of J14.
 - UART1_TX pin to TX pin of J14.
 - UART1_RX pin to RX pin of J14.
 - USB_D- pin to D- pin of J14.
 - USB_D+ pin to D+ pin of J14.
 - PWM1 pin to PWM1 pin of J14.
 - PWM2 pin to PWM2 pin of J14.
 - PWM3 pin to PWM3 pin of J14.
 - IO37 pin to TXD0 pin of J14.
 - IO36 pin to RXD0 pin of J14.
 - IO17 pin to UART1_TX pin of J14.
 - IO18 pin to UART1_RX pin of J14.
 - IO13 pin to USB_D- pin of J14.
 - IO14 pin to USB_D+ pin of J14.
 - IO21 pin to PWM1 pin of J14.
 - IO28 pin to PWM2 pin of J14.
 - IO29 pin to PWM3 pin of J14.
 - IO30 pin to IO37 pin of J14.
 - IO37 pin to TXD0 pin of J14.
 - IO38 pin to RXD0 pin of J14.
 - IO39 pin to UART1_TX pin of J14.
 - IO40 pin to UART1_RX pin of J14.
 - IO41 pin to USB_D- pin of J14.
 - IO42 pin to USB_D+ pin of J14.
 - IO45 pin to PWM1 pin of J14.
 - IO46 pin to PWM2 pin of J14.
 - IO47 pin to PWM3 pin of J14.
 - IO48 pin to IO48 pin of J14.
 - GND pin to GND.
- J14 USB_Micro**:
 - Shield pin to GND.
 - GND pin to GND.
 - D- pin to USB_D- pin of U4.
 - D+ pin to USB_D+ pin of U4.
 - VBUS pin to +3.3V supply.
- J15 FEM_HEADER_01x13**:
 - IO15 pin to TX pin of J14.
 - IO16 pin to RX pin of J14.
 - IO37 pin to TX pin of J14.
 - IO38 pin to RX pin of J14.
 - IO39 pin to TX pin of J14.
 - IO40 pin to RX pin of J14.
 - IO41 pin to TX pin of J14.
 - IO42 pin to RX pin of J14.
 - IO44 pin to TX pin of J14.
 - IO45 pin to RX pin of J14.
 - IO46 pin to TX pin of J14.
 - IO47 pin to RX pin of J14.
 - IO48 pin to TX pin of J14.

[illegible]

The diagram shows a component labeled J12 with part number 702460801. It has 8 pins. Pin 1 is labeled 'Vin' and is connected to a +12V source. Pin 2 is labeled 'Serial'. Pin 8 is labeled 'GND' and is connected to ground. Pins 3, 4, 5, 6, and 7 are marked with an 'X' and are not connected.

The diagram shows the J13 connector (702460801-AB) with the following connections:

- Pin 1:** Connected to +12V.
- Pin 2:** Connected to UART1_TX.
- Pin 3:** Connected to Vin.
- Pin 4:** Connected to Serial.
- Pin 5:** Connected to GND.
- Pin 6:** Connected to GND.
- Pin 7:** Connected to GND.
- Pin 8:** Connected to GND.

LED_PWR

D4
LED_RED

R5
66.5
0.25W

LED_MOV

D5
LED_ORANGE

R7
66.5
0.25W

LED_ROY

D6
LED_GREEN

R8
66.5
0.25W

LED_DBG1

D7
LED_YELLOW

R9
66.5
0.25W

LED_DBG2

D8
LED_GRN

R10
200
0.125W

Id: 1/1